

Python Programming

Unit 05 – Lecture 03: Inheritance and Types of Inheritance

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Repository: <https://github.com/tali7c/Python-Programming>

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Learning Outcomes

- Explain inheritance and its purpose

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- Create subclasses and reuse parent methods/attributes

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- Explain inheritance and its purpose
- Create subclasses and reuse parent methods/attributes
- Use `super()` to call parent constructor/methods
- Identify types of inheritance (single, multilevel, multiple, hierarchical)

What is Inheritance?

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- Reuse common code and extend behavior
- “is-a” relationship (Student is a Person)

Basic Syntax

```
class Person:
    def __init__(self, name):
        self.name = name

class Student(Person):
    def __init__(self, name, sapid):
        super().__init__(name)
        self.sapid = sapid
```

Types of Inheritance

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- Single: $A \rightarrow B$
- Multilevel: $A \rightarrow B \rightarrow C$
- Hierarchical: $A \rightarrow (B, C, D)$
- Multiple: $(A, B) \rightarrow C$

Why `super()`?

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- Calls parent class methods safely
- Avoids duplicating parent initialization code
- Important in multiple inheritance (method resolution order)

Demo: Person $\rightarrow$ Student $\rightarrow$ PlacementStudent

- File: `demo/inheritance_demo.py`

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- Demonstrates:
 - single and multilevel inheritance
 - use of `super()`

Checkpoint 1

Question: What does `super()` do in a subclass constructor?

Checkpoint 2

Question: Give one real-world example of inheritance.

Think-Pair-Share

Discuss:

- Should Car inherit from Engine? Why or why not?

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- Inheritance reuses and extends code
- Subclasses can add new attributes and methods
- `super()` helps call parent logic correctly
- Multiple inheritance exists, but use it carefully

Exit Question

What type of inheritance is: $A \rightarrow (B, C)$?